# STACKS, SHEAVES ON ALGEBRAIC STACKS: LEMMA-FIBRE-PRODUCTS

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 2863-2874 of sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex; proof span is lines 2876-2911, the complete author proof (34 body lines).

*Editorial note.* Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 13 of queue_algebraic-stacks-sheaves.json (pb=1841, stmt 2863-2874, proof 2876-2911).

*Editorial note.* Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-fibre-products' also exists in stacks-schemes — a different proposition in a different chapter file.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

*Source reference.* Original lemma 04T5: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2911–2911. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 04TB: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2911–2911. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 06N6: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2911–2911. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 03KP: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2911–2911. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 02ZB: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2911–2911. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let $p : \mathcal{X} \to (Sch/S)_{fppf}$ be a category fibred in groupoids.*

(1) *The category $\mathcal{X}$ has fibre products.*
(2) *If the Isom-presheaves of $\mathcal{X}$ are representable by algebraic spaces, then $\mathcal{X}$ has equalizers.*
(3) *If $\mathcal{X}$ is an algebraic stack (or more generally a quotient stack), then $\mathcal{X}$ has equalizers.*

*Proof.* Part (1) follows Categories, Lemma 06N6 as $(Sch/S)_{fppf}$ has fibre products.

Let $a, b : x \to y$ be morphisms of $\mathcal{X}$. Set $U = p(x)$ and $V = p(y)$. The category of schemes has equalizers hence we can let $W \to U$ be the equalizer of $p(a)$ and $p(b)$. Denote $c : z \to x$ a morphism of $\mathcal{X}$ lying over $W \to U$. The equalizer of $a$ and $b$, if it exists, is the equalizer of $a \circ c$ and $b \circ c$. Thus we may assume that $p(a) = p(b) = f : U \to V$. As $\mathcal{X}$ is fibred in groupoids, there exists a unique automorphism $i : x \to x$ in the fibre category of $\mathcal{X}$ over $U$ such that $a \circ i = b$. Again the equalizer of $a$ and $b$ is the equalizer of $\mathrm{id}_x$ and $i$. Recall that the $Isom_{\mathcal{X}}(x)$ is the presheaf on $(Sch/U)_{fppf}$ which to $T/U$ associates the set of automorphisms of $x|_T$ in the fibre category of $\mathcal{X}$ over $T$, see Stacks, Definition 02ZB. If $Isom_{\mathcal{X}}(x)$ is representable by an algebraic space $G \to U$, then we see that $\mathrm{id}_x$ and $i$ define morphisms $e, i : U \to G$ over $U$. Set $M = U \times_{e,G,i} U$, which by Morphisms of Spaces, Lemma 03KP is a scheme. Then it is clear that $x|_M \to x$ is the equalizer of the maps $\mathrm{id}_x$ and $i$ in $\mathcal{X}$. This proves (2).

If $\mathcal{X} = [U/R]$ for some groupoid in algebraic spaces $(U, R, s, t, c)$ over $S$, then the hypothesis of (2) holds by Bootstrap, Lemma 04TB. If $\mathcal{X}$ is an algebraic stack, then we can choose a presentation $[U/R] \cong \mathcal{X}$ by Algebraic Stacks, Lemma 04T5. □